Electrical Power and Energy Systems 173 (2025) 111403

Contents lists available at ScienceDirect

International Journal of Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes

Co-allocation and operational optimization for green power-direct-supply data center clusters with shared energy storage☆

Zeming Jiang [a], Yingzhao Tang [b,c,*], Shixing Ding [b,c], Yu Yang [a], Jingyi Zhao [a], Sicheng Tao [a], Yijun Xu [d]

[a] China United Network Communications Corporation Limited, Beijing 100033, China
[b] School of Electrical Engineering, Yanshan University, Qinhuangdao 066004, China
[c] Key Lab of Power Electronics for Energy Conservation and Motor Drive of Hebei Province (Yanshan University), Qinhuangdao 066004 Hebei Province, China
[d] School of Electrical Engineering, Southeast University, Nanjing 210096, China

ARTICLE INFO

Keywords:
Data center cluster
Shared energy storage
Robust optimization
Asymmetric Nash bargaining

ABSTRACT

The rapid proliferation of data center cluster (DCC) presents significant challenges to power system flexibility and renewable energy integration. This paper proposes a two-stage, bi-level collaborative framework to optimize the joint operation and capacity planning of DCC in Green Power Direct Supply (GPDS) mode and a Shared Energy Storage system. To manage uncertainties in renewable generation and workloads, a data-driven sample-based robust (DSRO) optimization approach is employed to mitigate the conservatism of traditional methods. In the first stage, a robust configuration model determines the optimal capacities of the SES and gas turbines, while linearizing the M/M/1 queuing model to guarantee a service violation rate below 5%. The second stage focuses on distributed operation. The complex, non-convex problem is decoupled into cost minimization and benefit allocation subproblems, solved efficiently using the alternating direction method of multipliers. Cooperative benefits are then distributed via an asymmetric Nash bargaining game, where a novel logarithmic function quantifies each participant's bargaining power based on their net energy contribution, ensuring fair allocation. Simulation results validate that the proposed framework significantly enhances system economy, renewable accommodation, and resource allocation flexibility, offering a robust foundation for multi-agent collaborative energy systems.

1. Introduction

Amidst the global convergence of energy transition and digital economy, greening data center clusters (DCC), as one of the critical energy-intensive infrastructure, is imperative. The Green Power Direct Supply (GPDS) model, integrating renewable energy (e.g., wind/solar) via physical/virtual lines, is pivotal for DCC to achieve carbon neutrality and reduce operational costs [1–3]. However, it introduces unprecedented challenges: a stark volatility-reliability contradiction between renewable energy's intermittency and DCC's stringent power supply requirements. To mitigate source-load fluctuations, shared energy storage (SES) systems—as flexible resources—can alleviate spatiotemporal energy mismatches through multi-agent sharing and collaborative optimization when integrated with GPDC-enabled DCC [4].

Traditional DCC's energy management often focuses on internal optimization of single entities or employs centralized control, failing to address the coupled challenges of multi-agent configuration and coordination. For such multi-agent, large-scale DCC, energy management transforms into a collaborative game-theoretic problem involving multiple stakeholders, complex interactions, and resource sharing [5]. Centralized strategies struggle to reconcile individual interests and privacy protection demands across DCC, resulting in low cooperation willingness or inefficiency [6]. Consequently, a critical challenge that emerges is how to fairly dispatch energy flows while maximizing overall benefits and ensuring each DCC's energy reliability and economic efficiency. Without effective coordination, this may cause imbalanced resource utilization, reduced system efficiency, or even cooperation breakdown. Crucially, these issues are accentuated in systems employing the GPDS model, whereby DCC faces heightened renewable energy fluctuations due to their direct coupling with wind and solar generation, while multi-agent dynamic interactions further compound optimization

☆ This article is part of a special issue entitled: 'DERs for Data Centers' published in International Journal of Electrical Power and Energy Systems.

* Corresponding author at: School of Electrical Engineering, Yanshan University, Qinhuangdao 066004, China.
E-mail address: tyz370502@163.com (Y. Tang).

https://doi.org/10.1016/j.ijepes.2025.111403
Received 6 July 2025; Received in revised form 16 August 2025; Accepted 12 November 2025
Available online 21 November 2025
0142-0615/© 2025 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Nomenclature

Abbreviations

CPE	Customer premise equipment
DC	Data Center
DCC	Data Center Cluster
GPDS	Green Power Direct Supply
EC	Electric chiller
RE	Renewable energy
SES	Shared energy storage
TPD	Task processing device

Indices

i	Index for DC
j	Index for batch workload
t	Time step index

Parameters

$S_{\mathrm{SES}}^{\mathrm{cap}}$	Planning capacity limit of SES
ρ_{SES}	Charge/discharge power limitation factor
$\overline{P}_i^{\mathrm{port}}$	Max/Min charge/discharge power of SES
$\eta_{\mathrm{ch/dis}}$	Charge/Discharge efficiency of SES
γ_{GT}	Heat recovery coefficient
γ_{COP}	Coefficient of Cooling Performance
λ_{cool}	Cold-electricity conversion coefficient
k_λ	Job arrival rate conversion rate coefficient
k_μ	Job service rate conversion rate coefficient
$L_{i,t}^0$	Baseline load
$P_{i,t}^{\mathrm{int}}$	Interactive workload
π_t^{buy}	Electricity purchase price
π_t^{sell}	Electricity sell price
π_{loss}	Energy storage degradation cost coefficient
π_{cur}	Discarding wind and solar cost coefficient
π_{gas}	Gas cost coefficient
π_{cool}	Heat recovery cooling cost coefficient
λ_{batch}	Penalty coefficient for the unfinished task j
∂_{SES}	The proportion of the scheduling period within the planned lifespan of the GT
∂_{GT}	The proportion of the scheduling period within the planned lifespan of the GT
$P_{\mathrm{grid}}^{\mathrm{max}}$	Max interactive power with the power grid
$P_{\mathrm{grid}}^{\mathrm{min}}$	Min interactive power with the power grid
$\varsigma_{\mathrm{deg}}^{\mathrm{SES}}$	SES degradation cost coefficient
$\varsigma_{\mathrm{ec}}^{\mathrm{GE}}$	Equipment depreciation cost factor

Variables

$S_{\mathrm{SES}}^{\mathrm{max}}$	Max capacity limit of SES
$P_{i,t}^{\mathrm{ch/dis}}$	Charge/discharge power of SES
$v_{i,t}^{\mathrm{ch/dis}}$	State of charge discharge of SES
$E_{\mathrm{SES},t}$	Rated energy capacity of SES
$P_{\mathrm{GT},i}^{\mathrm{max}}$	Max output power of GT
$P_{\mathrm{GT},i,t}$	Output electric power of GT
$Q_{\mathrm{GT},i,t}$	Output thermal power of GT
$Q_{\mathrm{EC},i,t}$	Cooling provided by electric chiller
$P_{\mathrm{EC},i,t}$	Electric power consumed by chiller
$P_{\mathrm{PV},i,t}^{\mathrm{forecast}}$	PV forecast power
$P_{\mathrm{PV},i,t}$	PV generation power
$P_{\mathrm{PV},i,t}^{\mathrm{cur}}$	PV curtailment
$P_{\mathrm{WT},i,t}^{\mathrm{forecast}}$	WT forecast power
$P_{\mathrm{WT},i,t}$	WT generation power
$P_{\mathrm{WT},i,t}^{\mathrm{cur}}$	WT curtailment
$y_{i,j,t}$	Flag indicating whether the task has started
$P_{\mathrm{batch},i,j,t}$	Batch workload
$E_{\mathrm{batch},i,j}$	Total batch workload
$D_{i,j}$	Processing time for a single batch workload
$Q_{\mathrm{batch},i,j,t}$	Thermal from batch workload processing
$\rho_{i,t}^{util}$	Resource utilization
$\lambda_{i,t}^{jobs}$	Job arrival rate
$\mu_{i,t}^{jobs}$	Job service rate
$\Delta L_{i,t}$	Workload adjustment
$L_{q,i,t}^{jobs}$	Job queue length
$W_{q,i,t}^{time}$	Assignment average waiting time
$t_{dly,i}^{peak/all}$	Average response delay of peak/all hours
$R_{\mathrm{exp}/con,i}^{\mathrm{max}}$	Maximum resource expansion/ contraction rate
$L_{i,t}^{final}$	Total final IT load processed by DC
$P_t^{\mathrm{buy},i}$	Power purchase from grid by DC
$P_t^{\mathrm{sell,SES}}$	Power sell from grid by SES
P_t^{buy}	Power purchase from grid by DCC
P_t^{sell}	Power sell from grid by DCC

complexity.

Addressing the aforementioned issues first requires resolving the coupled issues among energy supply management, equipment capacity configuration (such as energy storage equipment), and coordinated operation scheduling between computing workloads and energy systems in DCC. Reference [7] designed a detailed energy consumption model for the data center (DC), focusing on high-power and cooling demands, yet overlooked interactions with external equipment. Reference [8] optimized internal energy device configurations within DCC to minimize investment costs, but relied on deterministic forecasts of load and renewable energy (RE), failing to account for the dynamic operational uncertainties. Such models lack robustness against RE fluctuations and load variability under GPDS-enabled DCC scenarios, leading to suboptimal configurations in practice. In contrast, reference [9] primarily focused on short-term operation scheduling of DCC under fixed equipment configuration, optimizing the operation strategies of generators, energy storage systems, and controllable loads to reduce operational costs or RE curtailment rates. However, its scheduling flexibility is constrained by predetermined configurations, preventing full utilization of equipment potential and causing resource redundancy or inefficient operation. Particularly in GPDS-enabled DCC scenarios, the operation scheduling as operation scheduling may not be adapted to configuration schemes that consider RE fluctuations. To address this gap, some studies attempt to co-optimize configuration and scheduling. Reference [10] proposed a two-stage stochastic optimization method, where equipment capacity is decided in stage 1 and operation scheduling in stage 2, addressing RE uncertainty. Nevertheless, its centralized optimization introduces high computational complexity and cannot satisfy privacy requirements in multi-DCC collaborations.

The expansion of DCC and increased penetration of directly integrated green power exacerbate resource constraints within individual DCC. For such configured systems, how to handle multi-agent collaboration and fair revenue allocation remains a challenge [11]. Reference [12] explored energy sharing mechanisms among multiple microgrids, achieving total system cost reduction through centralized optimization. However, this centralized approach faces challenges in DCC because each DC, as an independent interest entity, places extreme importance on the privacy protection of its operational data. To address privacy

concerns and encourage participation from all parties, distributed optimization methods that share only partial information have gained widespread attention. Reference [13] employed a deviation correction mechanism to protect DC privacy, yet it lacked rigorous mathematical proof for the distribution of benefits among parties. Reference [14] utilized the alternating direction method of multipliers (ADMM) to achieve social welfare maximization in multi-microgrid distributed energy management, effectively protecting DC privacy and equally distributing reduced operational costs among all entities. This allocation method is unfair to entities that contribute more to the system, especially in GPDS-enabled DCC, where some DCs possess superior green power resources or more flexible loads. How to reflect their unique contributions in revenue allocation is central to stimulating cooperative potential. Consequently, reference [15] adopted an asymmetric Nash bargaining model to enhance fairness allocation among all parties.

Furthermore, Energy Storage systems enhance economic efficiency, reliability, and renewable utilization by providing flexible energy buffering services to DC. Reference [16] configured energy storage within DCs to improve system flexibility. However, decentralized storage configuration across DCC would increase costs significantly. Consequently, SES was introduced as a third-party service provider in Reference [17], offering flexible energy buffering and scheduling services for multiple DCs. Notably, the joint optimization of SES and DCC requires an in-depth investigation into fair synergistic benefit allocation, which is critical to incentivize participation from both DCC and SES operators. Although Reference [1] recognized the role of SES in coordinating DCC energy interactions, it omitted game-theoretic analysis of multi-agent interest relationships. The model in reference [18] adopted the Stackelberg game, failing to fully consider the dynamic negotiation process during the distributed operation of each agent.

Characterized by reliance on intermittent RE sources, a core feature of the GPDS model, DCC operational stability and economic costs are exposed to perturbations from multiple uncertainties, including wind turbine (WT) and photovoltaic (PV) output, computational workloads, and random contingencies [19]. Coordinated mitigation of renewable energy volatility and DCC workload stochasticity is therefore critical. Renewable energy output is influenced by weather conditions, exhibiting high intermittency and volatility, making precise prediction difficult [20]. Simultaneously, DCC workloads are also dynamic and stochastic, with arrival rates and processing demands varying over time. These uncertainties pose significant risks to the system's optimal configuration and operation scheduling [21]. Reference [22] proposed a robust optimization-based DC scheduling strategy, effectively addressing renewable energy uncertainty, but its conservatism limited economic benefits and could lead to an underestimation of green power consumption capacity. In recent years, data-driven robust optimization methods have gradually emerged, aiming to construct uncertainty sets using historical data to ensure robustness while reducing conservatism [23]. Reference [24] applied data-driven methods to model uncertainties in DCs, but few studies have applied them to the fine-grained energy management of DCC, especially in scenarios combining heterogeneous workloads within DCC and quality of service (QoS) guarantees. Therefore, a more refined and adaptive uncertainty handling method is needed to effectively reduce conservatism and improve economic efficiency while ensuring system reliability and QoS.

The internal workloads of DCC are diverse, and effective management of these workloads directly relates to the QoS provided by DCC [25]. For instance, by adjusting the processing time of batch workloads, they can be shifted to periods with abundant renewable energy output or lower electricity prices, thereby reducing operational costs. Reference [26] investigated the impact of internal load scheduling within DCC on energy consumption, but its models typically treat IT load as a whole, failing to finely distinguish the characteristics of different task types. Furthermore, the core function of DCC is to provide high-quality computing services; therefore, when performing energy optimization, QoS must be strictly guaranteed. Reference [27] attempted to incorporate DCC QoS into energy optimization models, but often employed simplified linear constraints, failing to fully capture the nonlinear characteristics of queuing systems within DCC. Queuing theory can more accurately describe DCC system utilization and average number of queued jobs, but its nonlinear characteristics introduce computational complexity to solving optimization models [28]. How to effectively integrate these nonlinear characteristics into optimization models to ensure cost optimization using green power without compromising core service quality is key to achieving fine-grained management and QoS guarantees within DCC.

Building on the research gap analysis, we observe that few works have systematically addressed the interrelated complex problems in GPDS-enabled DCC and SES within a unified framework, especially considering DCC heterogeneity, SES as an independent service role, renewable energy, DCC workload stochasticity handling, and nonlinear QoS constraints within DCC. To address the aforementioned challenges, this paper proposes an innovative two-stage, bi-level collaborative management framework aimed at optimizing the joint operation and capacity configuration of GPDS-enabled DCC and SES. Through an ingenious hierarchical design, this framework effectively decouples long-term investment planning from short-term operation scheduling and achieves multi-agent synergistic benefits and fair revenue allocation through optimization algorithms and game theory tools. The core objective of this paper is to significantly enhance system economic efficiency, renewable energy consumption rates, and resource configuration flexibility, providing theoretical and methodological support for future multi-agent collaborative energy systems.

In summary, the contributions of this paper are as follows:

(1) A two-stage, bi-level collaborative management framework is systematically constructed to address the challenging problem of optimal DCC configuration. The first stage focuses on configuration considering uncertainty, while the second stage addresses the distributed operation of data center clusters in robust scenarios after configuration.

(2) To achieve synergistic flexibility resource utilization within GPDS-enabled DCC, an integrated architecture is proposed where SES acts as an independent service entity. An asymmetric Nash bargaining mechanism is introduced to coordinate the interests of multiple data centers based on each system's contribution to the cluster, achieving privacy-preserving collaborative optimization through ADMM.

(3) To overcome the impact of high-proportion renewable energy uncertainty, inherent in GPDS, on stable data center power supply, data-driven sample robust methods are employed to characterize uncertainty. Furthermore, the distinct output characteristics of photovoltaic and wind power are leveraged to achieve complementary advantages within heterogeneous data center clusters.

The remainder of this paper is organized as follows. Section 2 presents the system architecture and the proposed operational framework. Section 3 details the mathematical models of the bi-level system components. Section 4 formulates the complete two-stage optimization problem, from robust configuration to operational scheduling via Nash bargaining. Section 5 describes the solution methodology, including the robust approach and linearization techniques. Section 6 provides case studies to validate the model's effectiveness, followed by the conclusion in Section 7.

2. System structure and management method

2.1. System structure

Fig. 1 illustrates the collaborative system architecture of the DCC and SES proposed in this paper. This system primarily comprises two core entities: the Shared Energy Storage Operator (SESO) and a heterogeneous DCC.

The SES is positioned at the upper layer of the system, with its investment and operation managed by an independent SESO. The SESO

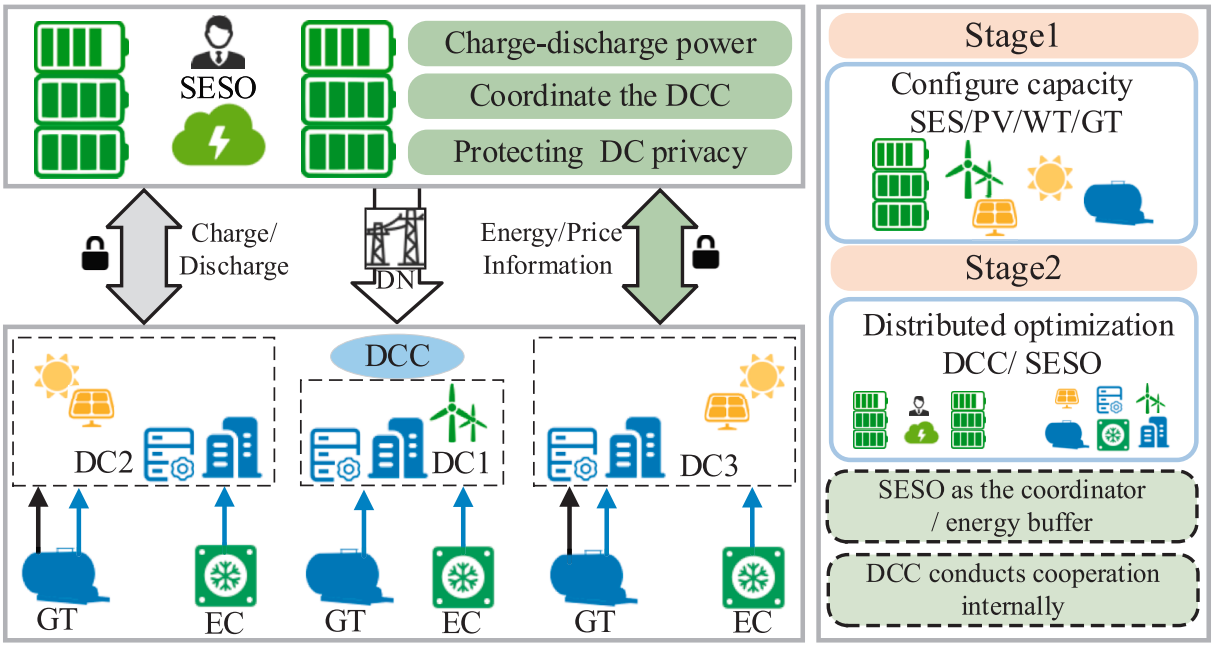


Fig. 1. Two-stage double-level collaborative optimization architecture diagram.

does not directly engage in the computing operations of data centers; instead, it functions as a third-party service provider, offering flexible energy charging and discharging services to the entire DCC by managing its large-capacity energy storage facilities. Its core functions include responding to the DCC's energy demands, coordinating the overall energy balance of the cluster, and safeguarding the operational privacy of individual data centers through secure information exchange channels. The DCC resides in the lower layer of the system and is composed of multiple DCs, each with distinct resource endowments. As depicted, each DC is equipped with fundamental computing power devices and cooling infrastructure, including servers and Electric chillers (EC), and is configured with gas turbines (GT) to serve as backup or supplementary power sources. DCs are connected to various RE generation equipment via tie lines, forming a heterogeneous energy cluster configuration with direct green power integration.

2.2. Operation management method

To efficiently manage the aforementioned system, this paper proposes a two-stage, bi-level collaborative optimization framework, as shown on the right side of Fig. 1. This framework hierarchically decouples the complex problems of long-term planning and short-term operation [28].

The Robust capacity configuration stage (Stage 1) is the upper-level, long-term investment decision-making stage. Its core objective is to determine the optimal capacities of key energy equipment at the beginning of the planning horizon, considering multiple uncertainties such as future renewable energy output, electricity prices, and computing loads, to minimize the total system cost. This includes the rated capacity and power of the SES, the installed capacity of PV and WT within each DC, and the rated power of GTs. Decisions made in this stage lay the physical foundation and cost boundaries for subsequent system operation.

The distributed collaborative operation (Stage 2) is the lower-level, short-term operational scheduling stage, building upon the equipment capacities determined in the first stage. This stage employs a bi-level optimization model to simulate the collaborative operation and game process between the DCC and SESO:

1) The SESO acts as a coordinator and energy buffer. Based on aggregated information obtained from the lower layer, the SESO formulates optimal charging and discharging strategies, aiming to maximize its profit or guide the system towards global optimality.

2) Each DC within the DCC, as an independent entity, optimizes the operational strategies of its internal energy equipment and flexible loads to minimize its operational costs, upon receiving coordination signals such as prices and dispatch commands from the SESO. Simultaneously, cooperation within the DCC facilitates internal energy exchange to maximize the local consumption of renewable energy.

This bi-level structure, while ensuring the decision-making independence and privacy of each entity, achieves closed-loop feedback and collaborative optimization between the upper and lower layers through parameter transmission, such as price signals and dispatch commands. The two-stage framework effectively addresses the coupling problem between investment decisions and operational scheduling, making the model more aligned with practical engineering applications.

3. Two-level model of DCC

3.1. Upper-level SES model

The SES system plays a crucial role in enhancing the flexibility, reliability, and economic efficiency of data center clusters. As a coordinator and energy buffer for these clusters, its optimal management is of paramount importance [29]. The upper-level model defines the key operational and physical constraints of the SES, which serve as the foundation for its capacity configuration and dispatch.$(\forall i,\ t)$, $\mathscr{D} \in \{\mathrm{DC1}, \mathrm{DC2}, \mathrm{DC3}\}$

The capacity limit constraint is as follows:

$$0 \le S_{\mathrm{SES}}^{\max} \le S_{\mathrm{SES}}^{\mathrm{cap}} \tag{1}$$

Total charging/discharging power constraints are as follows:

$$\begin{cases} \sum_{i\in\mathscr{D}} P_{i,t}^{\mathrm{ch}} \le \rho_{\mathrm{SES}} S_{\mathrm{SES}}^{\max} \\ \sum_{i\in\mathscr{D}} P_{i,t}^{\mathrm{dis}} \le \rho_{\mathrm{SES}} S_{\mathrm{SES}}^{\max} \end{cases} \tag{2}$$

SES power limit constraints are as follows:

$$\begin{cases} P_{i,t}^{\mathrm{ch}} \le v_{i,t}^{\mathrm{ch}} \cdot \overline{P}_i^{\mathrm{port}} \\ P_{i,t}^{\mathrm{dis}} \le v_{i,t}^{\mathrm{dis}} \cdot \overline{P}_i^{\mathrm{port}} \\ v_{i,t}^{\mathrm{ch}} + v_{i,t}^{\mathrm{dis}} \le 1 \\ v_{i,t}^{\mathrm{ch}}, v_{i,t}^{\mathrm{dis}} \in \{0,1\} \end{cases} \tag{3}$$

State of charge constraints are as follows:

$$\begin{cases} 0.1 S_{\mathrm{SES}}^{\max} \le E_{\mathrm{SES},t} \le 0.9 S_{\mathrm{SES}}^{\max} \\ E_{\mathrm{SES},t} = E_{\mathrm{SES},t-1} + \eta_{\mathrm{ch}} \sum_{i \in \mathscr{D}} P_{i,t}^{\mathrm{ch}} - \dfrac{\sum_{i \in \mathscr{D}} P_{i,t}^{\mathrm{dis}}}{\eta_{\mathrm{dis}}} \quad (\forall t \ge 2) \\ E_{\mathrm{SES},24} = 0.5 S_{\mathrm{SES}}^{\max} \end{cases} \tag{4}$$

3.2. Lower-level DCC model

Each DC operates as an independent entity within the lower-level problem, managing its internal energy resources and loads to minimize its operational costs. Data centers are unique in their diverse energy demands, encompassing Information Technology (IT) devices, cooling systems, and auxiliary services, as well as their potential for on-site power generation and load flexibility.$(\forall i,j,t)$

GT constraints are as follows:

$$\begin{cases} 0 \le P_{\mathrm{GT},i,t} \le P_{\mathrm{GT},i}^{\max} \\ 0 \le Q_{\mathrm{GT},i,t} \le \gamma_{\mathrm{GT}} \cdot P_{\mathrm{GT},i,t} \end{cases} \tag{5}$$

EC constraint is as follows:

$$P_{\mathrm{EC},i,t} = \gamma_{\mathrm{COP}} \cdot Q_{\mathrm{EC},i,t} \tag{6}$$

RE consumption constraints are as follows:

$$\begin{cases} P_{\mathrm{PV},i,t}^{\mathrm{forecast}} = P_{\mathrm{PV},i,t} + P_{\mathrm{PV},i,t}^{\mathrm{cur}} \\ P_{\mathrm{WT},i,t}^{\mathrm{forecast}} = P_{\mathrm{Wind},i,t} + P_{\mathrm{WT},i,t}^{\mathrm{cur}} \end{cases} \tag{7}$$

Arriving workloads at the DC can be classified into interactive and batch workloads based on their varying latency tolerance levels. Interactive workloads $P_{i,t}^{\mathrm{int}}$ typically involve real-time request processing, have high requirements for response time, and lack temporal flexibility. Batch workloads include tasks such as data analysis, machine learning training, and log processing, but have lower requirements for response time and can be flexibly scheduled over a longer time frame [13]. The DCC adopts a power-driven computing model, where it can dynamically adjust the processing of batch workloads $P_{\mathrm{batch},i,j,t}$ within a certain range based on the real-time availability of renewable energy. Both types of workloads are issued by CPE and processed by TPD. Firstly, interactive workloads are latency-sensitive and must therefore be processed immediately, while batch workloads offer temporal flexibility, which will be represented by a separate model below. The schematic diagram for batch loads is shown in Fig. 2, and the specific formulas are as follows (j represents the batch number for batch workloads):

$$\begin{cases} \sum_{\tau_j^{\min}}^{\tau_j^{\max}} y_{i,j,t} = 1 \\ P_{\mathrm{batch},i,j,t} = \dfrac{E_{\mathrm{batch},i,j}}{D_{i,j}} \sum_{\tau=\max(1,t-D_j+1)}^{t} y_{i,j,t} \\ Q_{\mathrm{batch},i,j,t} = \lambda_{\mathrm{cool}} P_{\mathrm{batch},i,j,t} \end{cases} \tag{8}$$

The QoS requirements for the DC are modeled based on M/M/1 queuing theory. In the data center context, IT load can be abstracted as "jobs" that need to be processed, and the computing resources of the data center provide "services". The formulas are as follows:

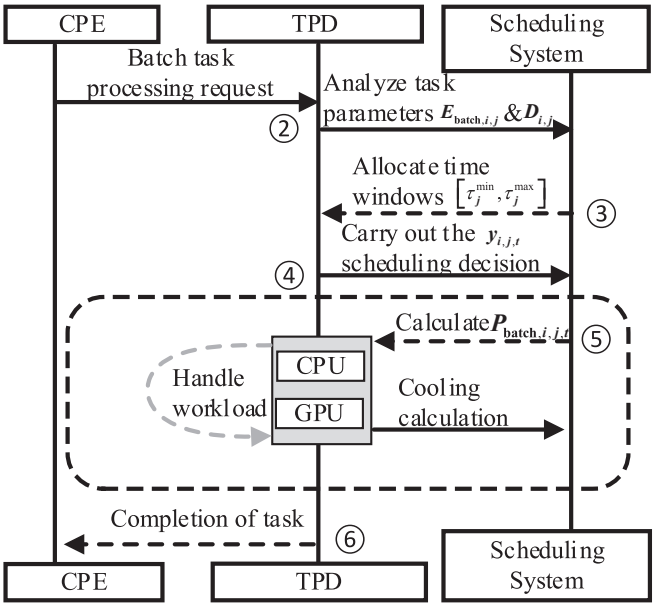


Fig. 2. Diagram of DC handling batch workload.

$$\begin{cases} \rho_{i,t}^{util} = \dfrac{\lambda_{i,t}^{jobs}}{\mu_{i,t}^{jobs}} \\ \lambda_{i,t}^{jobs} = k_{\lambda} \cdot L_{i,t}^{0} \\ \mu_{i,t}^{jobs} = k_{\mu} \cdot \left(L_{i,t}^{0} - \Delta L_{i,t} \right) \\ L_{q,i,t}^{jobs} = \dfrac{(\rho_{i,t}^{util})^2}{1 - \rho_{i,t}^{util}} \\ W_{q,i,t}^{time} = \dfrac{L_{q,i,t}^{jobs}}{\lambda_{i,t}^{jobs}} \end{cases} \tag{9}$$

Following the M/M/1 model, we further derive key QoS metrics for the data center, including response delay and resource expansion speed, as formulated below:

$$\begin{cases} t_{dly,i}^{peak} = \dfrac{1}{|\mathscr{T}_{peak}|} \sum_{t \in \mathscr{T}_{peak}} W_{q,i,t}^{time} \\ t_{dly,i}^{all} = \dfrac{1}{24} \sum_{t}^{24} W_{q,i,t}^{time} \end{cases} \tag{10}$$

$$\begin{cases} R_{\mathrm{exp},i}^{\max} = \max\limits_{t \in \{2,\ldots,24\}} (L_{i,t}^{final} - L_{i,t-1}^{final}) \\ R_{con,i}^{\max} = \min\limits_{t \in \{2,\ldots,24\}} (L_{i,t}^{final} - L_{i,t-1}^{final}) \end{cases} \tag{11}$$

All IT equipment within the DC, such as servers, storage devices, and network switches, consumes electrical energy during operation. Nearly all of this consumed electrical energy is ultimately converted into heat, which is dissipated into the machine room environment. To maintain a stable operating temperature inside the data center, the cooling capacity generated by the refrigeration system must be equal to the total heat generated by IT equipment and other auxiliary facilities. The cold energy balance constraint ensures that all optimization results are based on guaranteeing a safe operating temperature for servers, which is the foundation of the model's effectiveness. The specific formula is as follows:

$$Q_{\mathrm{EC},i,t} + Q_{\mathrm{GT},i,t} = \lambda_{\mathrm{cool}} P_{i,t}^{\mathrm{int}} + Q_{\mathrm{batch},i,t} \tag{12}$$

Electrical power balance constraints are as follows:

$$\left(P_{\mathrm{PV},i,t}-P^{\mathrm{cur}}_{\mathrm{PV},i,t}+P_{\mathrm{WT},i,t}-P^{\mathrm{cur}}_{\mathrm{WT},i,t}\right)+P_{\mathrm{GT},i,t}+P^{\mathrm{buy},i}_{t}+P^{\mathrm{dis}}_{i,t} = P^{\mathrm{int}}_{i,t}+\sum\nolimits_{j}P^{\mathrm{batch}}_{i,j,t}+P^{\mathrm{ch}}_{i,t}+P^{\mathrm{sell},i}_{t}+P_{\mathrm{EC},i,t} \tag{13}$$

4. Two-Stage configuration optimization method

The proposed framework employs a two-stage optimization approach to address different planning and operational aspects of data center clusters and shared energy storage systems. The first stage focuses on cost-driven robust capacity configuration, while the second stage deals with optimal operational scheduling and fair profit allocation through a sophisticated Nash bargaining model.

4.1. Stage 1 robust configuration model

The first stage of the optimization framework focuses on long-term investment decisions for the capacities of shared energy storage and distributed gas turbines. This stage is formulated as a robust optimization problem to address the inherent uncertainties in renewable energy generation and data center load. It ensures that the configured capacity can accommodate fluctuations in renewable energy and load, while simultaneously minimizing the total system cost.$(\forall i,j,t)$

The objective function is as follows:

$$\min_{\vec{u}\in\mathscr{U}} F^{U}=\begin{pmatrix}\partial_{\mathrm{SES}}\cdot C_{\mathrm{SES}}\cdot S^{\max}_{\mathrm{SES}}+\sum\limits_{i\in\mathscr{D}}\left(\partial_{\mathrm{GT}}\cdot C_{\mathrm{GT}}\cdot P^{\max}_{\mathrm{GT},i}\right)+\sum\limits_{t=1}^{T}\left(P^{\mathrm{buy}}_{t}\cdot\pi^{\mathrm{buy}}_{t}-P^{\mathrm{sell}}_{t}\cdot\pi^{\mathrm{sell}}_{t}\right)+\\ \sum\limits_{i\in\mathscr{D}}\sum\limits_{t=1}^{T}\pi_{\mathrm{loss}}\left(P^{\mathrm{ch}}_{i,t}+P^{\mathrm{dis}}_{i,t}\right)+\sum\limits_{i\in\mathscr{D}}\sum\limits_{t=1}^{T}\pi_{\mathrm{cur}}\left(P^{\mathrm{cur}}_{\mathrm{PV},i,t}+P^{\mathrm{cur}}_{\mathrm{WT},i,t}\right)+\sum\limits_{t=1}^{T}\left(\pi_{gas}\cdot P_{\mathrm{GT},i,t}+\pi_{\mathrm{cool}}\cdot Q_{\mathrm{GT},i,t}+\sum\limits_{j}\lambda_{\mathrm{batch}}\cdot(1-y_{i,j})\right)\end{pmatrix} \tag{14}$$

where C_{SES} is the investment cost per unit energy capacity of SES, C_{GT} is the investment cost per unit energy capacity of GT.

The constraints include (1–11) and the following additional constraints:

The SES capacity configuration constraint is as follows:

$$S^{\max}_{\mathrm{SES}}=\partial_{\mathrm{SES}}\max\left(P^{\mathrm{ch}}_{i,t}\right) \tag{15}$$

GT capacity configuration constraint is as follows:

$$P^{\max}_{\mathrm{GT},i}=\partial_{\mathrm{GT}}\max\left(P_{\mathrm{GT},i,t}\right) \tag{16}$$

Grid interaction constraints are as follows:

$$P^{\min}_{\mathrm{grid}}\le P^{\mathrm{buy}}_{t}=\sum_{i\in\mathscr{D}}P^{\mathrm{buy},i}_{t}\le P^{\max}_{\mathrm{grid}} \tag{17}$$

$$P^{\min}_{\mathrm{grid}}\le P^{\mathrm{sell}}_{t}=P^{\mathrm{sell,SES}}_{t}\le P^{\max}_{\mathrm{grid}} \tag{18}$$

Additionally, the system only permits DCC to purchase electricity and SES to sell electricity.

Global electrical power balance constraints are as follows:

$$P^{\mathrm{buy}}_{t}+\sum_{i}\left(P_{\mathrm{PV},i,t}-P^{\mathrm{cur}}_{\mathrm{PV},i,t}+P_{\mathrm{WT},i,t}-P^{\mathrm{cur}}_{\mathrm{WT},i,t}\right)+\sum_{i\in\mathscr{D}}P^{\mathrm{dis}}_{i,t}+\sum_{i}P_{\mathrm{GT},i,t} = \sum_{i}P_{\mathrm{EC},i,t}+P^{\mathrm{sell}}_{t}+\sum_{i}P^{\mathrm{int}}_{i,t}+\sum_{i\in\mathscr{D}}P^{\mathrm{ch}}_{i,t}+\sum_{i}\sum_{j}P^{\mathrm{batch}}_{i,j,t} \tag{19}$$

4.2. Stage 2 operation model

The second stage of the optimization framework focuses on optimizing the system's operational scheduling under the configured capacities determined in the first stage. This stage aims to minimize the operational costs of data centers and maximize the operational revenue of the shared energy storage by optimizing energy generation, consumption, and the energy exchange between the shared energy storage and individual data centers.

The SES system model is as follows:

$$\max_{\vec{u}\in\mathscr{U}} f^{L}_{\mathrm{SES}}=C^{\mathrm{SES}}_{\mathrm{trade}}-C^{\mathrm{SES}}_{\mathrm{deg}} \tag{20}$$

$$C^{\mathrm{SES}}_{\mathrm{trade}}=\pi^{\mathrm{sell}}_{t}\sum_{i\in\mathscr{D}}\sum_{t=1}^{T}P^{\mathrm{sell,SES}}_{t} \tag{21}$$

$$C^{\mathrm{SES}}_{\mathrm{deg}}=\zeta^{\mathrm{SES}}_{\mathrm{deg}}\sum_{t=1}^{T}\left(P^{\mathrm{ch}}_{i,t}+P^{\mathrm{dis}}_{i,t}\right) \tag{22}$$

where $C^{\mathrm{SES}}_{\mathrm{trade}}$ represents the revenue from SES electricity sales, and $C^{\mathrm{SES}}_{\mathrm{deg}}$ represents the degradation cost of the energy storage system. The constraints are Equations. (5–13) and (17,18).

The model for PV-type DGCP DC is as follows:

$$\min_{\vec{u}\in\mathscr{U}} f^{L}_{\mathrm{PDC}}=C^{\mathrm{GE}}_{\mathrm{ec}}+C^{\mathrm{PV}}_{\mathrm{op}} \tag{23}$$

$$C^{\mathrm{GE}}_{\mathrm{ec}}=\zeta^{\mathrm{GE}}_{\mathrm{ec}}\sum_{t}\left(P_{\mathrm{EC},i,t}+P_{\mathrm{GT},i,t}\right) \tag{24}$$

$$C^{\mathrm{PV}}_{\mathrm{op}}=\sum_{t}\pi^{\mathrm{buy}}_{t}P^{\mathrm{buy}}_{t}-\sum_{t}\pi^{\mathrm{sell}}_{t}P^{\mathrm{sell}}_{t}+\pi_{\mathrm{cur}}\sum_{t}Q^{\mathrm{cur}}_{\mathrm{PV},t}+\pi_{gas}\sum_{t}P_{\mathrm{GT},t} \tag{25}$$

where $C^{\mathrm{PV}}_{\mathrm{op}}$ is the operation cost of PV-type DGCP DC, $C^{\mathrm{GE}}_{\mathrm{ec}}$ is the depreciation cost of GT and EC. The constraints are Equations. (5–13) and (17,18).

The model for WT-type DGCP DC is as follows:

$$\min_{\vec{u}\in\mathscr{U}} f^{L}_{\mathrm{WDC}}=C^{\mathrm{GE}}_{\mathrm{ec}}+C^{\mathrm{WT}}_{\mathrm{op}} \tag{26}$$

$$C^{\mathrm{WT}}_{\mathrm{op}}=\sum_{t}\pi^{\mathrm{buy}}_{t}P^{\mathrm{buy}}_{t}-\sum_{t}\pi^{\mathrm{sell}}_{t}P^{\mathrm{sell}}_{t}+\pi_{\mathrm{cur}}\sum_{t}Q^{\mathrm{cur}}_{\mathrm{WT},t}+c_{gas}\sum_{t}P_{\mathrm{GT},t} \tag{27}$$

where $C^{\mathrm{WT}}_{\mathrm{op}}$ is the operation cost of WT-type DGCP DC. The constraints are (5–11) and (15).

4.3. Asymmetric Nash bargaining model

This chapter first establishes a cooperative game model for data center clusters and shared energy storage based on Nash bargaining theory, and then promotes the fair and just distribution of cooperative benefits between data center clusters and shared energy storage systems through asymmetric bargaining.

4.3.1. The basic theory of Nash bargaining

The Nash bargaining optimization model investigated in this paper is a cooperative game[30]. After first maximizing the collective interests of

all participants, the cooperative gains are distributed among multiple participants through mutual negotiation. The Nash bargaining model satisfies a set of axioms, including symmetry, Pareto optimality, independence of irrelevant alternatives, and scale invariance. The standard Nash bargaining model is shown in Equation (28), and the solution that maximizes the Nash product is the equilibrium solution to the Nash bargaining game problem. $\mathscr{I} = \{\mathrm{SES}, \mathrm{DC1}, \mathrm{DC2}, \mathrm{DC3}\}$, $(\forall i \in \mathscr{I})$

$$\max \prod_{i \in \mathscr{I}} \left(C_i^0 - C_i^{\mathrm{co}}\right)^{d_i} \tag{28}$$

$$s.t.\ C_i^{\mathrm{co}} \le C_i^0 \tag{29}$$

where C_i^{co} represents the cost after coalition cooperation, and C_i^0 represents the cost before coalition cooperation, i.e., the disagreement point. The Nash bargaining model in this study is used to determine the optimal allocation of cost savings achieved through the joint operation of data center clusters and shared energy storage. This model aims to achieve a solution that is both efficient and fair, while considering each participant's individual contribution and bargaining power.

The Nash bargaining model, as shown in Equation (28), is a non-convex, non-linear problem with multiple coupled variables. Therefore, the aforementioned model is decomposed and transformed into two subproblems: the cost minimization subproblem for the data center cluster and shared energy storage coalition (P1), and the revenue allocation subproblem (P2), which are solved sequentially.

4.3.2. Solution to the cost minimization sub-problem (P1)

Maximizing coalition benefits is equivalent to minimizing costs. The saved costs represent the benefits achieved through cooperation. The total cost is the sum of the energy costs of each data center. The coalition cooperation and sharing model, constructed based on the Nash bargaining theory model is as follows:

$$\min\left(C^{\mathrm{SES}} + \sum_{i \in \mathscr{D}}^{3} C_i^{\mathrm{DC}}\right) \tag{30}$$

where C^{SES} represents the optimized cost for SES, and C_i^{DC} represents the optimized cost for the i-th DC. When solving subproblem P1, the internal electricity transaction cost within the coalition $C_i^{\mathrm{P2P}} = 0$. This is because P1 solely focuses on minimizing the total cost of the coalition, without considering how internal payments are settled. Internal payments are handled in subproblem P2, which deals with the allocation of benefits.

Therefore, for P1, the net internal payment balance is effectively zero. Since the energy sharing balance constraint (17) is multiply coupled among all microgrids, auxiliary variable vectors $P_{i\to\mathrm{ESS},t}$ and $P_{\mathrm{ESS}\to i,t}$ are introduced to transform problem P1 into a doubly coupled model.

$$P_{i\to\mathrm{ESS},t} + P_{\mathrm{ESS}\to i,t} = 0 \quad \forall i, t \tag{31}$$

This structure allows for the application of ADMM, a method highly effective for solving separable convex optimization problems with coupled linear constraints. ADMM decomposes the global problem into two locally solvable subproblems, coordinated through Lagrange multipliers. The augmented Lagrangian function $\mathscr{L}^{\mathrm{P1}}$ for P1 is constructed as follows:

$$\mathscr{L}^{\mathrm{P1}} = C^{\mathrm{SES}} + \sum_{i \in \mathscr{D}}^{3} C_i^{\mathrm{DC}} + \sum_{i=1}^{3}\sum_{t=1}^{T} \lambda_{i,t}\left(P_{i\to\mathrm{ESS},t} + P_{\mathrm{ESS}\to i,t}\right) + \frac{\rho}{2}\sum_{i=1}^{3} \times \sum_{t=1}^{T} \left\| P_{i\to\mathrm{ESS},t} + P_{\mathrm{ESS}\to i,t} \right\|^2 \tag{32}$$

where $\lambda_{i,t}$ is the Lagrange multiplier, and ρ is the penalty term coefficient for P1. The shared energy storage and each data center update their trading power strategies through local computation, with only trading power information exchanged between entities. Let k denote the iteration number. In each iteration, the following steps are performed:

SES update

$$\left(P_{\mathrm{ESS}\to i,t}^{(k+1)}\right) = \mathrm{argmin}\mathscr{L}_\rho\left(P_{\mathrm{ESS}\to i,t}, P_{i\to\mathrm{ESS},t}^{(k)}, \lambda_{i,t}^{(k)}\right) \tag{33}$$

DCC update

$$\left(P_{i\to\mathrm{ESS},t}^{(k+1)}\right) = \mathrm{argmin}\mathscr{L}_\rho\left(P_{\mathrm{ESS}\to i,t}^{(k+1)}, P_{i\to\mathrm{ESS},t}, \lambda_{i,t}^{(k)}\right) \tag{34}$$

Lagrange multiplier update

$$\lambda_{i,t}^{(k+1)} = \lambda_{i,t}^{(k)} + \rho\left(P_{i\to\mathrm{ESS},t}^{(k+1)} + P_{\mathrm{ESS}\to i,t}^{(k+1)}\right) \tag{35}$$

After all primal variables are updated, the Lagrange multipliers are updated based on the violation of coupled constraints. This update pushes the primal variables towards a consistent solution that satisfies the coupled constraints, and then these updated multipliers are passed to all participants for the next iteration.

4.3.3. Solution to the sub-problem of revenue distribution based on asymmetric bargaining (P2)

Once the optimal coalition operation and minimum total cost are determined in P1, the next crucial step is to fairly distribute the cooperative benefits between the data centers and the shared energy storage. This is achieved through P2, a subproblem that employs an asymmetric Nash bargaining model.

In P2P energy sharing scenarios, both providing and receiving energy are considered contributions, as they reduce reliance on the main grid. However, the ability to provide energy is generally valued more than the ability to receive energy, primarily because the cost of purchasing electricity from the main grid is typically higher than the price at which surplus energy can be sold back to the grid. This inherent asymmetry in energy valuation justifies the asymmetric bargaining approach.

The concept of asymmetric bargaining originates from economics, referring to situations where negotiating parties possess different bargaining powers due to asymmetries in information or status during the bargaining process. This paper adopts a nonlinear function based on the natural logarithm to quantify the contribution size of different microgrids in energy sharing. Subsequently, microgrids negotiate with each other, using their respective contributions as bargaining power, to determine their P2P electricity transaction prices and fairly distribute the benefits of energy sharing. Specifically, this is formulated as follows:

First, the total energy provided E_i^{sup} and energy received E_i^{rev} by each microgrid within the optimization time horizon are calculated. Subsequently, a nonlinear energy mapping function, formed by an exponential function with the natural constant 'e' as its base, is used to quantify each microgrid's bargaining power d_i based on its contribution to energy sharing:

$$d_i = \exp\left(\frac{E_i^{\mathrm{sup}}}{E_{\max}^{\mathrm{sup}}}\right) - \exp\left(-\frac{E_i^{\mathrm{rev}}}{E_{\max}^{\mathrm{rev}}}\right) \tag{36}$$

$$\begin{cases} E_i^{\mathrm{sup}} = \sum_t \max(0, P_{i\to\mathrm{ESS},t}) \\ E_i^{\mathrm{rev}} = -\sum_t \min(0, P_{i\to\mathrm{ESS},t}) \\ E_{\max}^{\mathrm{sup}} \max_i\left(E_i^{\mathrm{sup}}\right) \\ E_{\max}^{\mathrm{rev}} \max_i\left(E_i^{\mathrm{rev}}\right) \end{cases} \tag{37}$$

Finally, based on the established bargaining power quantification model, the optimal interactive power obtained from subproblem P1 is substituted into subproblem P2. The asymmetric bargaining benefit distribution model for the data center cluster and shared energy storage, constructed based on the Nash bargaining model, is shown in the

following equation:

$$\max\left[(C^0_{\mathrm{SES}} - C^*_{\mathrm{SES}} + \Delta C^{\mathrm{P2P}}_{\mathrm{SES}})\cdot \prod_{i=1}^{3}(C^0_i - C^*_i + \Delta C^{\mathrm{P2P}}_i)^{d_i}\right] \tag{38}$$

$$\begin{aligned} s.t.\ &\sum_{i\in\mathscr{I}} \Delta\, C^{\mathrm{P2P}}_i = 0 \\ C^0_i - C^*_i + \Delta\, C^{\mathrm{P2P}}_i &> 0 \quad \forall i \end{aligned} \tag{39}$$

where C^0_{SES} is the initial cost of SES, C^*_{SES} is the cost after P1 optimization by SES, $\Delta C^{\mathrm{P2P}}_{\mathrm{SES}}$ represents the revenue increase achieved through P2 optimization, C^0_i is the cost after P1 optimization by DC, C^*_i is the cost after P1 optimization by DC, $\Delta C^{\mathrm{P2P}}_i$ represents the revenue increase achieved through P2 optimization. The natural logarithm is a strictly monotonically increasing convex function. Therefore, by taking the logarithm of the above equation, the problem of finding the maximum value can be transformed into the problem of finding the minimum value, which is easier to solve. After the transformation, the following equation is obtained:

$$\min\left[-\ln(C^0_{\mathrm{SES}} - C^*_{\mathrm{SES}}) - d_i\sum_{i-1}^{3}\ln(C^0_i - C^*_i)\right] \tag{40}$$

Similar to P1, the ADMM algorithm is used to solve P2 in a distributed manner. The augmented Lagrangian function for subproblem P2 is constructed to handle coupled constraints and facilitate distributed updates. The augmented Lagrangian function $\mathscr{L}^{\mathrm{P2}}$ for P2 is established as

$$\begin{aligned} \mathscr{L}^{\mathrm{P2}} = \ & -\ln(C^0_{\mathrm{SES}} - C^*_{\mathrm{SES}} + \Delta C^{\mathrm{P2P}}_{\mathrm{SES}}) - \sum_{i=1}^{3} d_i \ln(C^0_i - C^*_i \\ & + \Delta C^{\mathrm{P2P}}_i) + \sum_i \mu_i \Delta C^{\mathrm{P2P}}_i + \frac{\sigma}{2}\Big\|\sum_i \Delta C^{\mathrm{P2P}}_i\Big\|^2 \end{aligned} \tag{41}$$

where ρ is the penalty term coefficient of P2, In each iteration, the following steps are performed:

SES and DC update

$$\Delta\, C^{\mathrm{P2P},(k+1)}_i = \mathrm{argmin}\mathscr{L}^{\mathrm{P2}} \tag{42}$$

Lagrange multiplier update

$$\mu^{(k+1)} = \mu^{(k)} + \sigma\sum_i \Delta\, C^{\mathrm{P2P},(k+1)}_i \tag{43}$$

The Lagrange multipliers are updated based on the violation of the total savings constraint, pushing the allocated savings towards the total cooperative surplus. These updated multipliers are then communicated to all participants for the next iteration.

5. Solution method

5.1. Data-driven sample robust optimization

Inherent uncertainties in renewable energy generation and data center workloads pose significant challenges for optimal capacity planning. Traditional robust optimization methods, while guaranteeing feasibility in worst-case scenarios, often lead to overly conservative and economically inefficient solutions. Data-driven sample robust optimization (DSRO) is an uncertainty optimization technique that combines robust optimization with data-driven approaches. It aims to address the issue of traditional robust optimization being overly conservative while utilizing historical data to more accurately characterize uncertainty [31]. Its core principle is to use historical observed data to construct an uncertainty set that can encompass the majority of observed samples/scenarios. This method, by introducing a confidence level, allows the optimization model to, to some extent, disregard a few extreme or anomalous historical samples. This, in turn, reduces the conservatism of decisions and improves economic efficiency, all while ensuring a certain level of reliability[32]. Furthermore, if the model is extended to a larger scale, DSRO can still maintain a good computational speed compared to SO.

For an optimization problem where certain constraints are affected by an uncertain parameter $\widetilde{u}$, given historical observed samples $u_s \in \mathbb{R}^k$, $s = 1,2,\ldots,n_o$ construct the uncertainty set $\mathscr{U} = \{\widetilde{u}|\exists S \subseteq \{1,\ldots,N\}, |S| \geq \lceil \eta N \rceil, \widetilde{u} = u_s \forall s \in S\}$, S represents the sample scenarios.

For the robust upper bound constraints L^{upper}_t, to determine the minimum configuration capacity required to meet most load demands:

$$\inf \sum L^{\mathrm{upper}}_t \tag{44}$$

$$L^{\mathrm{upper}}_t \geq u_{s,t} - M(1 - z_{s,t}) \tag{45}$$

$$\sum_{s=1}^{N} z_s \geq \lceil \eta N \rceil\ ,\ z_s \in \{0,1\} \tag{46}$$

For the robust lower bound constraints L^{lower}_t, to determine the minimum available output to ensure system reliability:

$$\sup \sum L^{\mathrm{lower}}_t \tag{47}$$

$$L^{\mathrm{lower}}_t \leq u_{s,t} + M(1 - z_{s,t}) \tag{48}$$

$$\sum_{s=1}^{N} z_s \geq \lceil \eta N \rceil\ ,\ z_s \in \{0,1\} \tag{49}$$

where $\eta \in [0,1]$ is the pre-defined confidence level. N represents the total number of historical observed samples. $\lceil \eta N \rceil$ s the ceiling value, ensuring that at least a proportion η of samples are selected for coverage. By introducing robust upper and lower bounds, the model can more comprehensively evaluate and manage uncertainty, ensuring the system operates under uncertainty in both computational load and renewable energy.

5.2. Tangential approximation method

Since the formulas for system utilization $\rho^{util}_{i,t}$ and the average number of queued jobs $L^{jobs}_{q,i,t}$ 的are nonlinear, they need to be linearized. The tangent approximation method is a common approach to transform nonlinear convex constraints into linear ones. Its core idea is that for a convex function $f(x)$, its tangent at any point x_k is always located below the function curve. Therefore, by selecting a sufficient number of tangent points within the function's domain and constructing tangents, these tangents can be used to approximate the original nonlinear function.

1)Tangent approximation for system utilization $\rho^{util}_{i,t}$: taking the first and second derivatives with respect to it $f\left(\mu^{jobs}_{i,t}\right)$, it is found that $f\left(\mu^{jobs}_{i,t}\right)$ is a convex function (given $f'\left(\mu^{jobs}_{i,t}\right) \geq 0$ and $f''\left(\mu^{jobs}_{i,t}\right) \geq 0$). For each DC i and time t, and the pre-selected N tangent points, $k \in \{1,2,\ldots,N\}$:

$$\rho^{util}_{i,t} \geq \frac{\lambda^{jobs}_{i,t}}{\mu_k} - \frac{\lambda^{jobs}_{i,t}}{\mu_i^2}(\mu^{jobs}_{i,t} - \mu_k) \tag{50}$$

where $0 \leq \rho^{util}_{it} \leq 1 - \epsilon^{util}$, ϵ^{util} is an extremely small positive number, ensuring $\rho^{util}_{it} < 1$.

2)Tangent approximation for average number of queued jobs: taking the first and second derivatives with respect to $g\left(\rho^{utd}_{i,t}\right)$, it is found that $g\left(\rho^{utd}_{i,t}\right)$ is a convex function (given $g'\left(\rho^{utd}_{i,t}\right) \geq 0$ and). For each DC i and time t, and the pre-selected tangent points, $k \in \{1,2,\ldots,N\}$:

$$L_{q,i,t}^{jobs} \geq \frac{\rho_k^2}{1-\rho_k} + \frac{2\rho_k - \rho_k^2}{(1-\rho_k)^2}(\rho_{i,t}^{util} - \rho_k) \tag{51}$$

At the same time, it is necessary to ensure that the number of queued jobs is non-negative, i.e., $L_{q,i,t}^{jobs} \geq 0$.

5.3. ADMM steps

To reasonably solve the asymmetric Nash bargaining problem, this chapter employs the ADMM algorithm to sequentially address two subproblems: cooperative revenue maximization and energy trading settlement. The basic principles of ADMM are detailed in reference [33]. The iterative steps and update schemes are formally presented below:

1)Initialization stage: Initialize the original variables $P_{ESS\to i,t}^{(0)} = 0$, $P_{i\to ESS,t}^{(0)} = 0$, $\forall i \in \{1,2,3\}, t \in T$, initialize the dual variables $\lambda_{i,t}^{(0)} = 0$, $\mu^{(0)} = 0$, set the penalty parameters $\rho = 0.01$, set the convergence accuracy threshold $\epsilon_{pri} = \epsilon_{dual} = 0.01$, initialize the iteration counter $k = 0$ and communicate with the data center cluster and shared energy storage to obtain the initial values of the original variables.

2) Alternating update of original variables: Update the optimization variables according to the formulas in 3.3.2 and 3.3.3.

3) Update of dual variables: Update the Lagrange multipliers according to the formulas in 3.3.2 and 3.3.3.

4) Convergence verification:

P1:Calculate the original residuals $r_{P1}^{(k+1)}$ and the dual residuals $s_{P1}^{(k+1)}$, conduct a convergence check:

$$\begin{cases} r_{P1}^{(k+1)} = \sum_t \sum_i \left\| P_{ESS\to i,t}^{(k+1)} + P_{i\to ESS,t}^{(k+1)} \right\|_2 \\ s_{P1}^{(k+1)} = \rho \sum_t \sum_i \left\| P_{ESS\to i,t}^{(k+1)} - P_{i\to ESS,t}^{(k+1)} \right\|_2 \end{cases} \tag{52}$$

If $r^{(k+1)} < \epsilon_{pri} \cap s^{(k+1)} < \epsilon_{dual}$, then it converges.

P2:Calculate the original residuals $r_{P2}^{(k+1)} r_{P1}^{(k+1)}$ and the dual residuals $s_{P2}^{(k+1)}$, conduct a convergence check:

$$\begin{cases} r_{P2}^{(k+1)} = \left\| \sum_i \Delta C_i^{P2P(k+1)} \right\|_2 \\ s_{P2}^{(k+1)} = \sigma \left\| \sum_i \left(\Delta C_i^{P2P(k+1)} - \Delta C_i^{P2P(k)} \right) \right\|_2 \end{cases} \tag{53}$$

If $r_{P2}^{(k+1)} < \epsilon_{pri} \cap s_{P2}^{(k+1)} < \epsilon_{dual}$, then it converges.

If these conditions are met, the algorithm for the revenue distribution sub-problem has converged, ensuring fair and optimal revenue

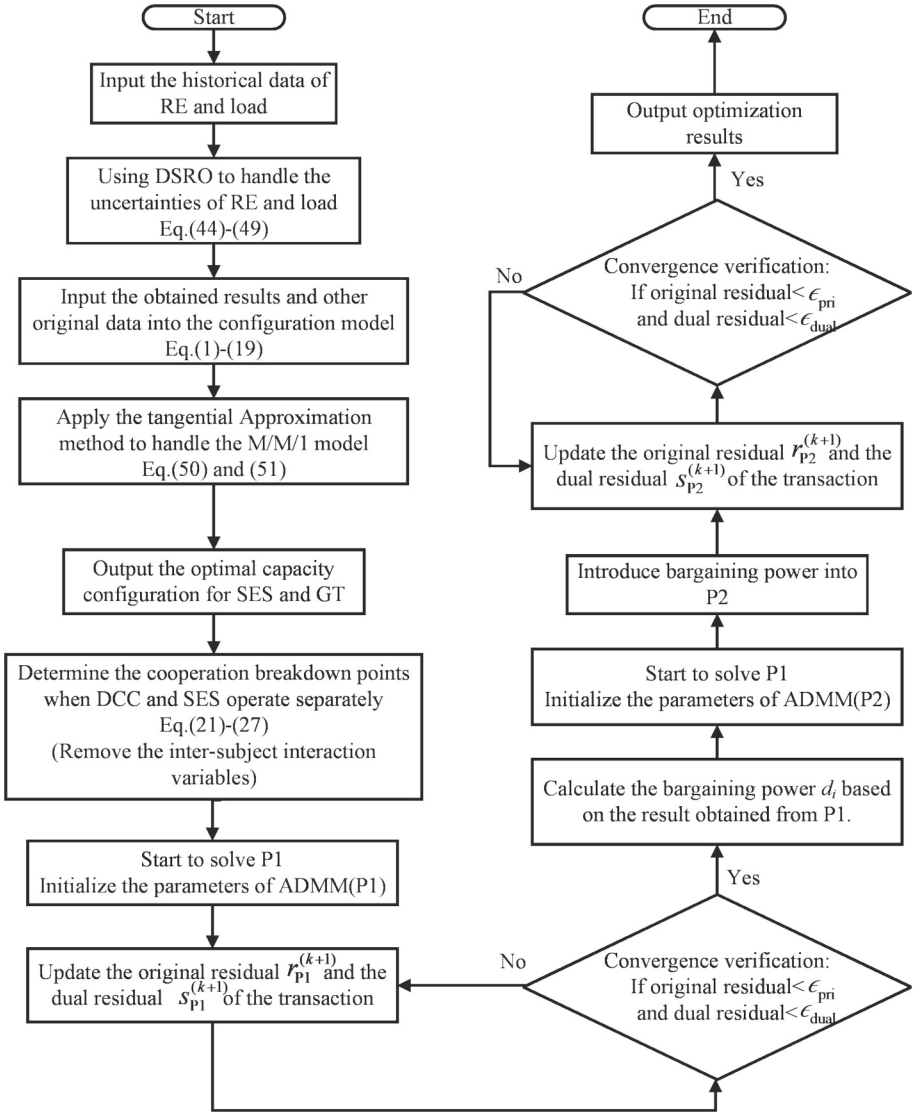
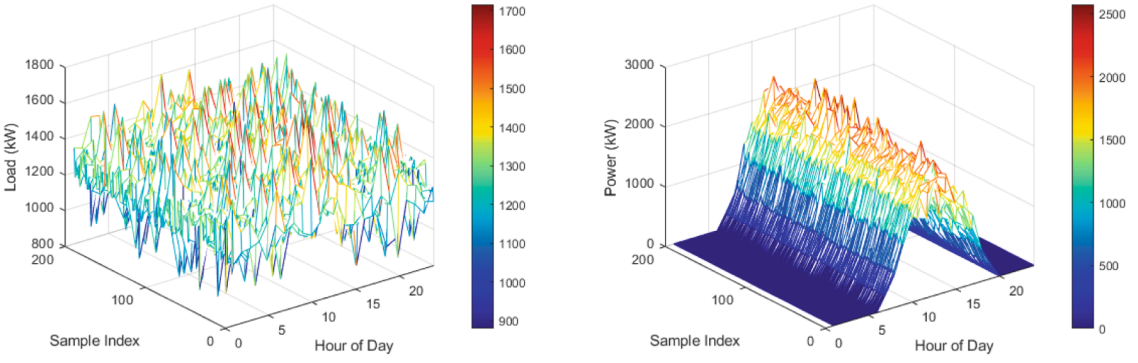
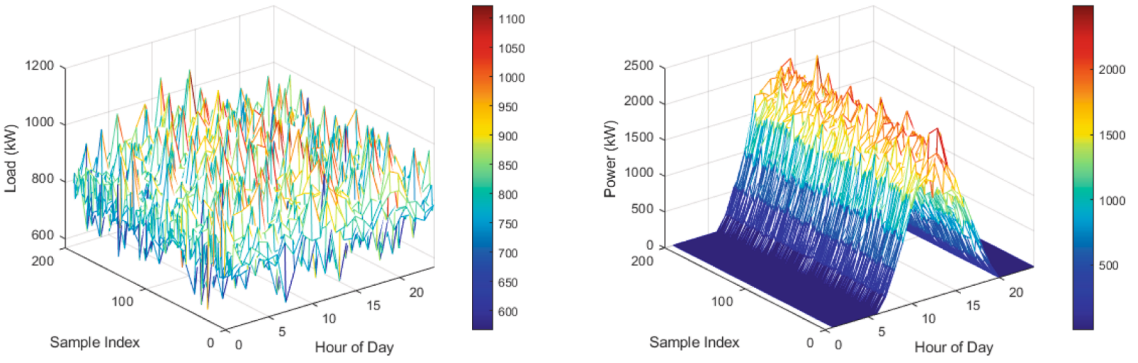


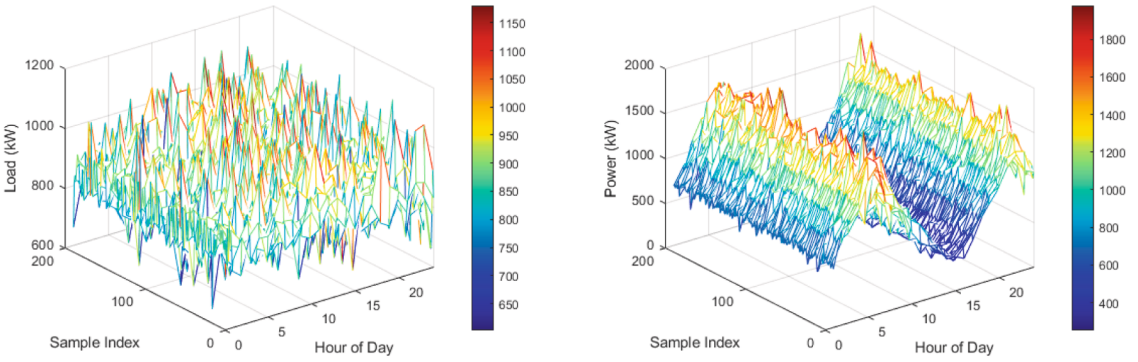
Fig. 3. Optimized flowchart.



(a) Sample of DC1 historical scene

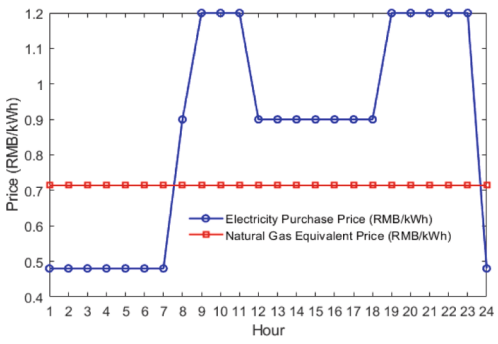


(b) Sample of DC2 historical scene

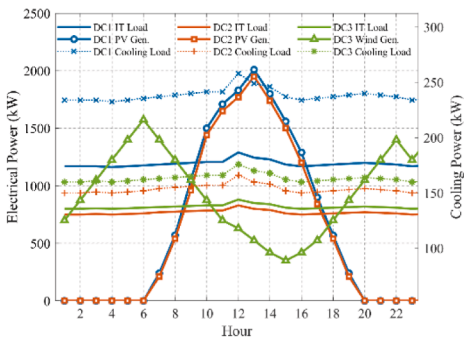


(c) Sample of DC3 historical scene

Fig. 4. Sample diagram of historical scenes in DCC history.



(a) Energy prices of DCC input/output



(b) Each DC load and RE output

Fig. 5. DCC parameters.

Table 1
Technical parameters of DCC and SES.

Equipment	Parameter	Value	Equipment	Parameter	Value
SES	η_{ch}	0.95	GT	η^{e}_{CHP}	0.35
	η_{dis}	0.95		γ_{GT}	0.5
	S^{cap}_{SES}(kWh)	5000		$\bar{P}^{port}_{i}$(kW)	1000
	∂_{SES}	5		∂_{GT})	5
EC	λ_{cool}	0.3	Grid	P^{max}_{grid}(kW)	500
	γ_{COP}	0.3		P^{min}_{grid}(kW)	0

distribution based on the asymmetric bargaining model.

The complete optimization process of the model is illustrated in Fig. 3.

6. Case study

6.1. Simulation system Description

To validate the effectiveness of the proposed framework, the paper selects a joint system consisting of a data center cluster with 3 heterogeneous data centers and shared energy storage for case studies. Fig. 4 shows the historical green power output and load samples of the DCC. Fig. 5(a) presents the electricity purchase and sale prices (i.e., the time-of-use electricity prices in the region) and natural gas prices of the joint system. Fig. 5(b) displays the electricity loads and renewable energy outputs of the three data centers. Table 1 shows the technical parameters of the equipment in the system, and Table 2 presents the cost coefficients for system operation. In addition, the Big M parameter is set to 10,000 to balance the efficiency and accuracy of model solving. Since the proposed sample-robust two-stage bi-level cooperative framework is finally reformulated as a solvable Mixed-Integer Linear Programming (MILP) model, it is directly solved using GUROBI in MATLAB, with a planning and scheduling interval of one hour.

6.2. Stage1 configuration results

In the configuration phase, two scenarios are established for comparative analysis of the proposed strategy's effectiveness. The scenario settings are as follows:

1)Scenario 1: Considers shared energy storage.

2)Scenario 2: Does not consider shared energy storage.

Through the robust optimization in the first stage of the two-stage framework, the optimal results are obtained, yielding the optimal capacity configurations for shared energy storage, gas turbines, and renewable energy units, as shown in Table 3.

It can be observed that the total cost decreased by 3915.14, representing a saving of 16.7 %. Although the investment cost was slightly higher due to the deployment of SES, the operating cost decreased by 4774.77, representing a reduction of 21.6 %, fully offsetting the increased investment cost and leading to significant net savings.

The deployment of SES acts as an energy buffer, enabling electricity to be purchased and stored during low electricity price periods and discharged during high electricity price periods. Consequently, this reduces the system's reliance on GTs, decreases the configured capacity of fossil energy supply equipment, and optimizes the energy structure.

Fig. 6 illustrates the electricity supply and demand within the DCC, showing that the DCC's full-day electricity demand can largely be met by renewable energy sources. Fig. 7 depicts the cooling power balance within a single DC, where the cooling power generated from GT waste heat can largely satisfy most of the demand. However, since DC2 is configured with a smaller GT capacity, the proportion of electric cooling is relatively higher.

Fig. 8 illustrates the comparison of GT electrical and thermal power configured in a single DC, while Fig. 9 shows the periods of workload processing in a single DC. Through the above comparison, we found that batch workloads tend to be processed during periods when renewable energy is abundant, as processing batch workloads is cheaper at these times.

According to the QoS service indicators in Fig. 10, it can be seen that both the service rate and arrival rate are low during 0–8o'clock and 18–24o'clock, indicating that the system is in a low-load state. From 9 to 17o'clock, the service rate significantly increases and exceeds the arrival rate, indicating that the system possesses excellent dynamic adjustment capabilities, allowing it to enhance service capacity during peak periods and avoid overload. From 9–17o'clock, especially from 11–14o'clock, the utilization rate drops to 0.6–0.7, indicating that the system has surplus processing capacity and has accelerated during peak hours. The decrease in system utilization is due to an increase in the service rate rather than a decrease in demand, which is a manifestation of proactive capacity expansion. During 9–17o'clock, both the average number of jobs currently waiting for service and the average waiting time for a job in the queue sharply decrease, indicating that the system responds quickly and has short waiting times. During peak periods, due to the improvement in service capacity, there are fewer queues and shorter waiting times, which indicates that the peak response mechanism is well-designed. Table 4 compares key QoS metrics, including response delay and maximum resource expansion/contraction rates, for intuitive illustration, while Fig. 11 presents a heatmap of workload migration.

6.3. Stage 2 asymmetric Nash-Bargaining results

Fig. 12(a) and Fig. 13(a) illustrate the iterative convergence of the ADMM algorithm employed in this paper to solve subproblem 1 and subproblem 2, respectively. Fig. 12(a) shows the convergence results of the power interaction between each DC and SES in subproblem 1. Fig. 12 (c) demonstrates the optimization effect on the total cost. This indicates that the proposed ADMM-based distributed solution algorithm, used for solving the subproblems of DC cost minimization and electricity trading

Table 2
Coefficients for calculating costs.

Equipment	GT/ EC				SES			DC	PV/WT
Coefficients(Yuan /kWh)	π_{gas}	π_{cool}		ς^{GE}_{ec}	π_{loss}	ς^{SES}_{deg}		λ_{batch}	π_{cur}
Value	0.714	0.1	0.005		0.005		0.005	0.2	0.5

Table 3
Optimization results after stage 1.

Case	Cost	Operating cost	Investment cost (converted to the current scheduling period)	Program results SES	GT1	GT2	GT3
Configured SES	19485.01	17348.24	2128.44	4727.50	429.76	277.20	295.20
Not configured SES	23400.15	22123.01	1277.14	–	656.03	490.38	305.92

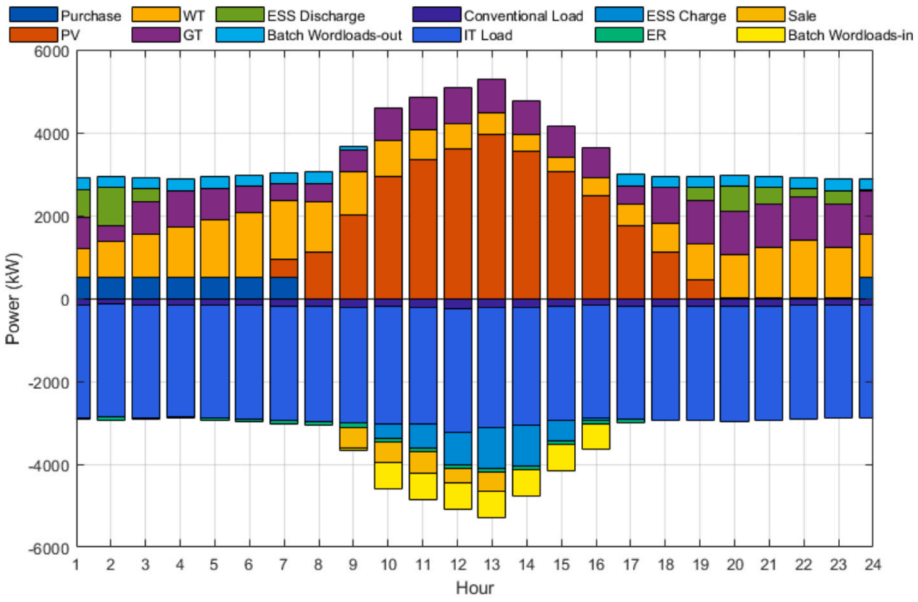
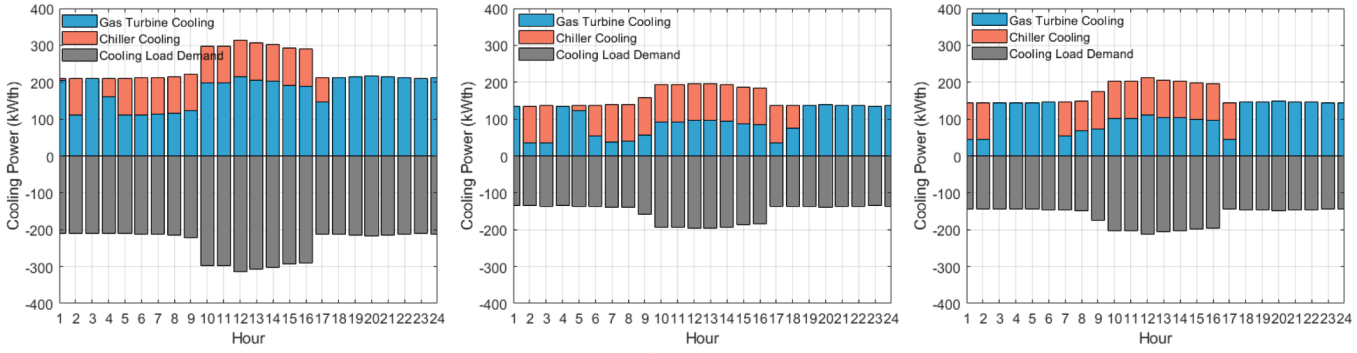
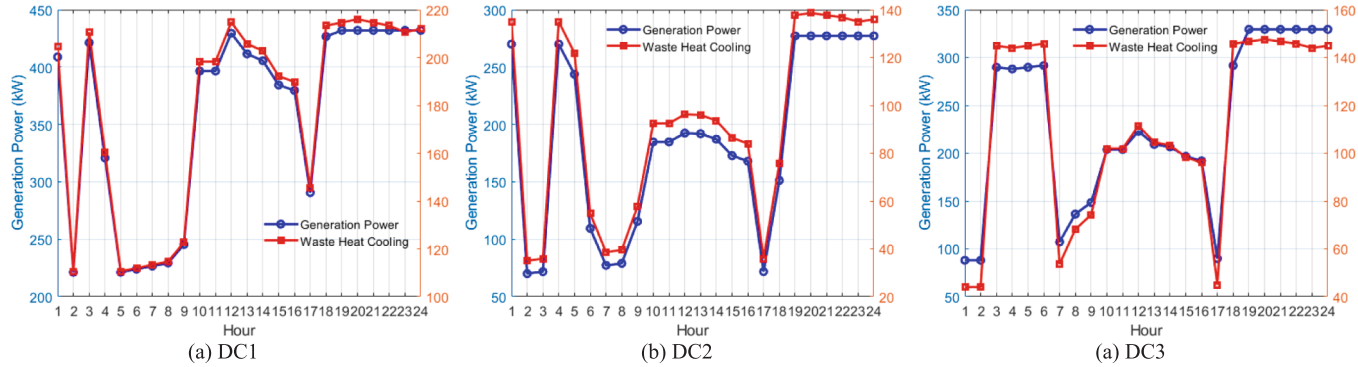


Fig. 6. Electric power balance diagram.



(a) DC1 (b) DC2 (a) DC3

Fig. 7. Cooling power balance.



(a) DC1 (b) DC2 (a) DC3

Fig. 8. Comparison of GT electric heating power.

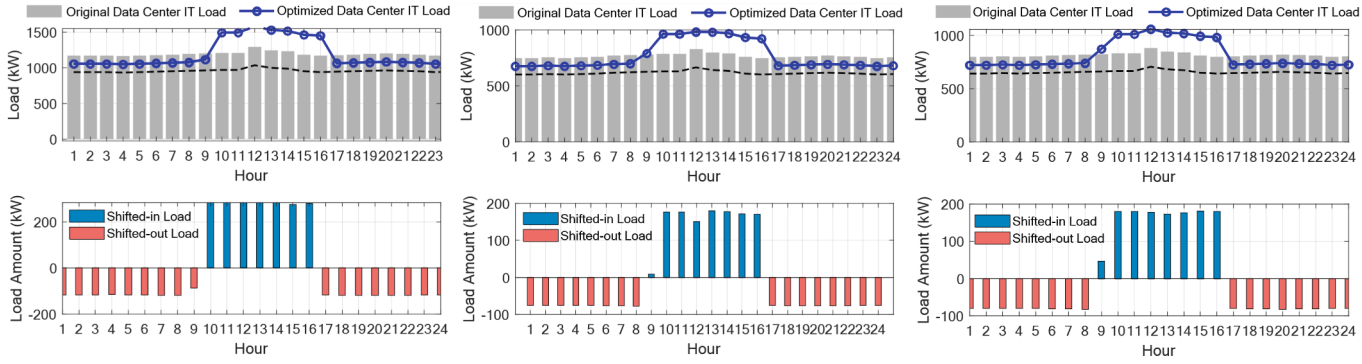
revenue maximization, exhibits good convergence performance and can ensure information privacy among the DCs.

To validate the effectiveness of the optimization method proposed in this paper, we set up the following scenarios for comparative analysis:

1)Scenario 3: SES coordinates the operation of DCs, and DCs work collaboratively, which is the optimized operation method described in this paper.

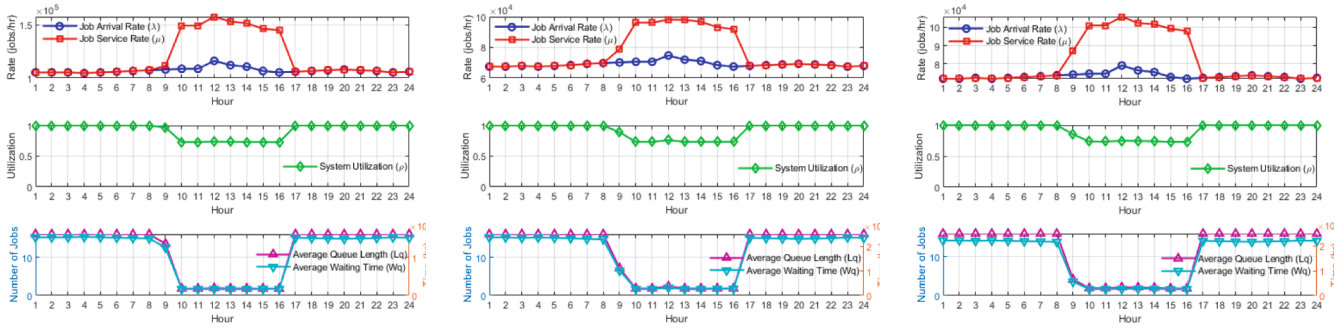
2)Scenario 4: SES does not coordinate the operation of DCs, and DCs operate independently, which is the disagreement point.

3)Scenario 5: SES does not coordinate the operation of DCs, but DCs work collaboratively, which is a cooperative game without SES participation.



(a) DC1 (b) DC2 (a) DC3

Fig. 9. Workload transfer situation.



(a) DC1 (b) DC2 (a) DC3

Fig. 10. Comparison of QoS services.

Table 4
QoS Indicators of DCC.

QoS Indicators	DC1	DC2	DC3
$t_{dly,i}^{peak}$(s)	0.157	0.209	0.187
$t_{dly,i}^{all}$(s)	0.398	0.605	0.564
$R_{exp,i}^{max}$(kW/h)	368.244	167.299	144.709
$R_{con,i}^{max}$(kW/h)	−394.047	−240.638	−255.397

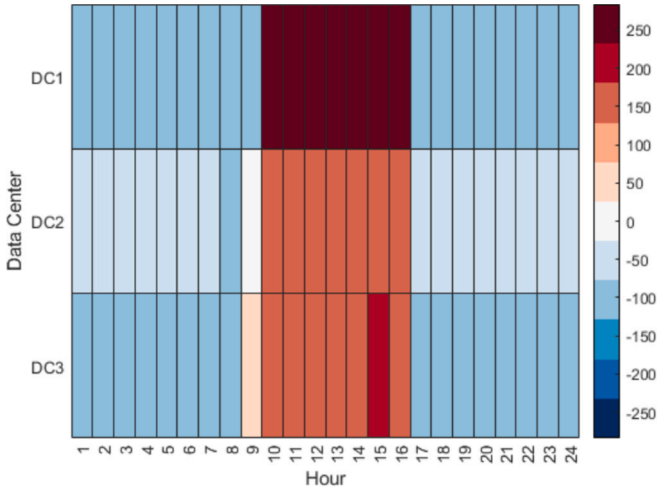


Fig. 11. Workload heatmap transfer.

The optimization results for each scenario are shown in Table 5.

Scenario 3 achieved the lowest total cost, demonstrating that the ADMM-based collaborative optimization method proposed in this paper, under SES coordination, can significantly reduce the operating costs of the entire data center cluster. Its operating cost is 21520.48 lower than that of the independent operation mode, and 37.3 % lower than that of the mode without shared energy storage.

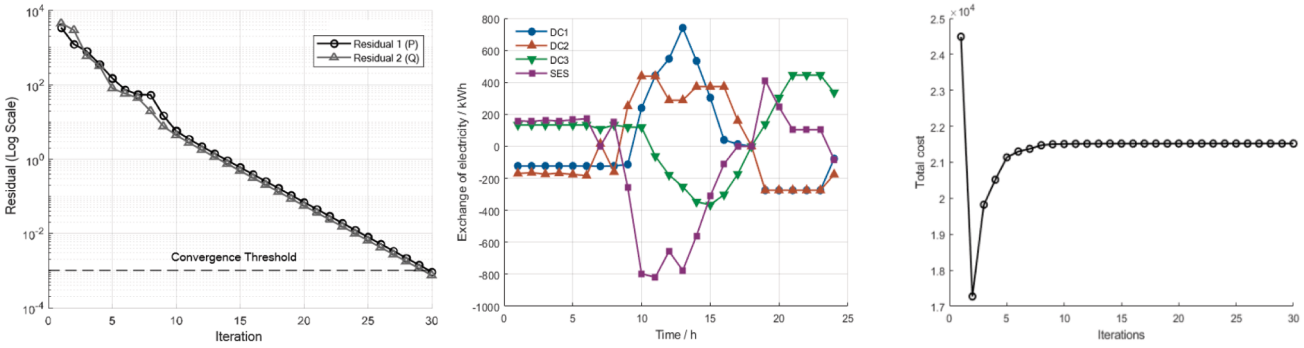
Scenario 4 had the highest total cost because, in this scenario, each DC operated independently and without SES coordination, leading to inefficient resource allocation and an inability to achieve overall optimality.

The total cost of Scenario 5 was between that of Scenario 3 and Scenario 4. Although DCs collaborated, the optimization effect was not as good as in Scenario 3 due to the lack of SES participation and coordination. This highlights the critical role of SES in promoting the overall system benefits.

The results indicate that both the collaborative operation among DCs and the coordinated participation of SES can effectively reduce the operating costs of individual data centers, achieving a win–win situation for all parties. Specifically, the coordinating.

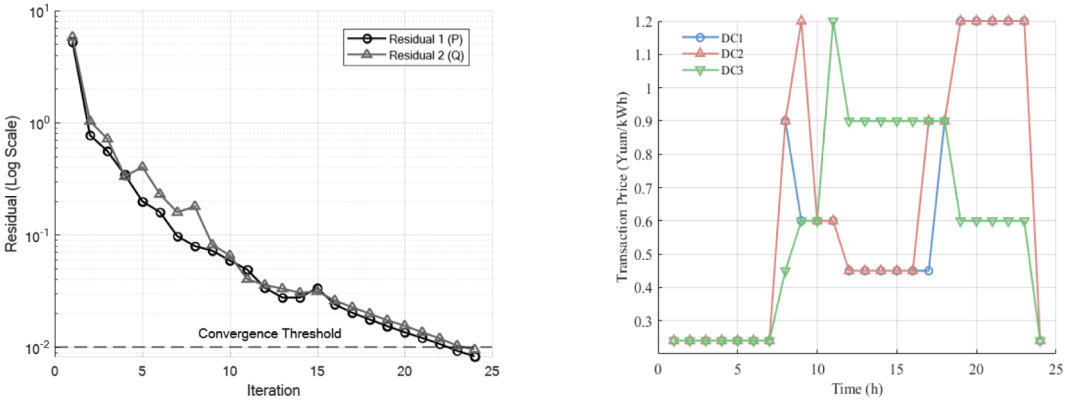
In the power interaction diagram, it can be observed that the two PV-type DGCP DCs tend to charge the SES during periods of abundant sunlight. This enables the WT-type DGCP DC to receive a significant amount of low-cost electricity during midday, leading to the SES extensively charging DC3 during this period. During nighttime hours, when PV output is zero, the WT-type DGCP DC primarily undertakes the task of charging the SES.

Fig. 13(b) presents the electricity price transaction diagram, illustrating how electricity prices vary over time during power transactions among the three data centers: DC1, DC2, and DC3. To highlight the price advantage of the trading model presented in this paper, the electricity



(a) The result of the iterative convergence of the P1 (b) Electric energy interaction result (b) Total cost iteration

Fig. 12. Optimization result of P1.



(a) The result of the iterative convergence of the P2 (b) Price transaction result

Fig. 13. Optimization result of P2.

Table 5
Optimization results after Nash bargaining.

Case	Scenario 3	Scenario 4	Scenario 5
Total Cost (Yuan)	21520.48	34326.12	25723.34
SES Earning (Yuan)	330.54	0	0
DC1 Cost (Yuan)	12963.58	16162.15	13218.71
DC2 Cost (Yuan)	5897.29	11338.39	8335.46
DC3 Cost (Yuan)	2329.06	6825.58	4169.17

price for inter-DC transactions is set higher than the price for selling electricity to the main grid. It can be observed that while the transaction electricity price curves for different data centers show similar trends, there are differences in specific values and the timing of peak occurrences. This may reflect variations in their respective electricity demands, supply capacities, or trading strategies.

6.4. DSRO sensitivity and performance analysis

The study conducted an uncertainty sensitivity analysis on the system's configuration results under different confidence levels to evaluate the impact of risk aversion on investment decisions and economic costs. The analysis results, as shown in Table 6, reveal a key strategic turning point. In the mid-to-high confidence interval of 70 % to 90 %, the system tends to adopt a balanced, synergistic configuration of Shared Energy Storage (SES) and Gas Turbines (GTs). It deploys large-capacity SES to mitigate volatility while equipping medium-capacity GTs as a supplementary resource. Furthermore, as the confidence level increases, the system shows a growing preference for larger GT capacities, reflecting the operational limits of SES in hedging against higher uncertainty. When the confidence level is raised to 100 %, corresponding to a strategy robust against the most extreme uncertainty scenarios, the optimal configuration undergoes a fundamental shift. The system completely forgoes the high-initial-investment energy storage, opting instead to significantly increase the installed capacity of each gas turbine. This phenomenon indicates that when faced with the risk of extreme energy shortages, the model deems on-demand generation from flexible GTs to be more cost-effective than energy time-shifting via storage. This strategic pivot is also reflected in the cost breakdown: although the total cost and operating cost rise with the confidence level, the investment cost at 100 % confidence drops markedly due to the abandonment of expensive storage assets.

To evaluate the computational efficiency of the proposed DSRO model in handling large-scale uncertainty, this study compared its

Table 6
Configuration results under different confidence levels.

Confidence	SES capacity	GT1 capacity	GT2 capacity	GT3 capacity	Total cost	Operating cost	Investment cost
70 %	4813.7	398.4	260.8	257.9	19258.5	2171.43	17089.92
80 %	4727.5	429.76	277.2	295.2	19485.01	17348.24	2128.44
90 %	4398.1	456.9	331.3	347.2	19951.03	2157.06	17781.95
100 %	0	773.8	694.5	711.7	20293.09	1818.42	18408.13

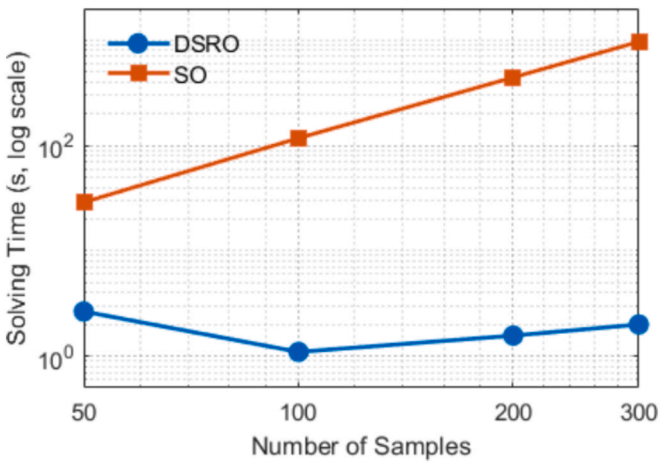


Fig. 14. Comparison chart of solving speeds of DSRO and SO.

Table 7
Computational Performance of DSRO and SO.

Number of Samples	DSRO Solving speed (s)	SO
50	2.6532	28.904
100	1.0945	117.52
200	1.5646	443.95
300	1.9963	967.8

solution speed with that of an SO model. The number of test scenarios was expanded from 50 to 300, with the results shown in Fig. 14 and Table 7. The computation time of the SO model exhibits explosive growth as the number of scenarios increases, climbing sharply from 28.9 s for 50 scenarios to 967.8 s for 300 scenarios, demonstrating its high sensitivity to the problem scale.

In contrast, the DSRO model, which employs a cardinality-constrained robust optimization framework, showcases superior computational performance and stability. This method directly uses data-driven scenarios to construct a discrete ambiguity set and introduces an adjustable confidence parameter to limit the number of worst-case scenarios the system must be robust against. By incorporating binary decision variables, the optimizer can endogenously select the most impactful scenarios from the entire set without requiring any a priori screening. This computational stability demonstrates the efficiency and scalability of the DSRO model in handling large-scale uncertainty problems.

7. Conclusion

This study addresses the collaborative optimization problem of direct green power integration with DCC and SES, proposing a set of innovative solutions. Firstly, this paper establishes a DCC model that integrates SES configuration and optimization. Building upon this, an internal energy coordination mechanism is developed, fully leveraging the complementary characteristics among different types of green electricity. Finally, based on this mechanism, the DCC can engage in cooperative games under the coordination of SES and distribute benefits according to their contribution to the overall system. Considering the uncertainty of renewable energy, a sample-robust optimization model is proposed for minimizing DCC costs and maximizing SES benefits.

The conclusions are as follows:

1)By constructing a two-stage bi-level collaborative management framework, the cross-time-scale coupling problem of long-term capacity configuration and short-term operational dispatch is effectively solved. Case studies demonstrate that SES, acting as an energy buffering unit, stores electricity during low-price periods and discharges during peak periods, reducing reliance on GTs and leading to a 35–47 % reduction in GT capacity investment for each DC.

2)Fair and efficient revenue distribution is achieved through an asymmetric Nash bargaining mechanism. The contribution-based game theory model successfully coordinates the interests of distributed entities, maximizing the absorption of green electricity. The cooperative game under SES coordination reduces total operating costs by 37.3 %. While SES gains a daily revenue of 330.54 yuan, the costs for each DCC decrease by 15–24 %, achieving a win–win situation for multiple parties.

3)Operational stability is ensured through a DSRO design, combining data-driven and robust optimization methods to address uncertainty. By covering renewable energy and load fluctuations with a 90 % confidence level, it helps prevent unnecessary increases in conservative costs. In scenarios with high proportions of renewable energy prediction errors, the QoS violation rate is controlled within 5 %.

Finally, the ADMM-based collaborative optimization primarily considers the energy layer, and scenarios involving workload allocation and scheduling for the DCC as a whole are outside the scope of this study. Moreover, the multi-agent collaborative game and revenue distribution mechanism envisioned by the model encounter complex barriers in terms of power market access and regulation. Future research will integrate server-level control algorithms, such as dynamic task assignment and feedback coordination mechanisms, to enhance the practicality of the model. Utilizing advanced data transmission encryption technology will also be a potential research hotspot.

CRediT authorship contribution statement

Zeming Jiang: Writing – review & editing, Funding acquisition, Formal analysis, Conceptualization. **Yingzhao Tang:** Writing – original draft, Software, Methodology. **Shixing Ding:** Project administration, Investigation, Data curation. **Yu Yang:** Resources. **Jingyi Zhao:** Supervision. **Sicheng Tao:** Validation. **Yijun Xu:** Software, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Smart Grid-National Science and Technology Major Project (Grant No. 2025ZD0804801).

Data availability

Data will be made available on request.

References

[1] Han O, Ding T, Zhang X, et al. A shared energy storage business model for data center clusters considering renewable energy uncertainties. Renew Energy 2023; 202:1273–90.
[2] Kwon S. Ensuring renewable energy utilization with quality of service guarantee for energy-efficient data center operations. Appl Energy 2020;276:115424.
[3] Zhou Y, Wang Y, Yang M. Optimal integration of renewable energy in refinery hydrogen management systems: Energy storage and direct utilization. Energ Conver Manage 2024;304:118223.
[4] Liu J, Chen X, Xiang Y, et al. Optimal planning and investment benefit analysis of shared energy storage for electricity retailers. Int J Electr Power Energy Syst 2021; 126:106561.
[5] Akbar S, Akbar UU, Ullah R, et al. A Coalitional Game-Based Adaptive Scheduler Leveraging Task Heterogeneity for Greener Data Centers. IEEE Trans Green Commun Networking 2024;9(1):55–69.
[6] Zheng K, Wang X, Liu J. Distributed traffic flow consolidation for power efficiency of large-scale data center network. IEEE Trans Cloud Comput 2020;10(2): 996–1007.

[7] Norani M, Deymi-Dashtebayaz M. Energy, exergy and exergoeconomic optimization of a proposed CCHP configuration under two different operating scenarios in a data center: Case study. J Clean Prod 2022;342:130971.
[8] Cheung H, Wang S. Optimal design of data center cooling systems concerning multi-chiller system configuration and component selection for energy-efficient operation and maximized free-cooling. Renew Energy 2019;143:1717–31.
[9] Wang J, Deng H, Liu Y, et al. Coordinated optimal scheduling of integrated energy system for data center based on computing load shifting. Energy 2023;267:126585.
[10] He W, Xu Q, Guo R, et al. Optimization analysis of different distributed energy configurations for data centers. Sustain Mater Technol 2024;41:e01005.
[11] Yang J, Xu W, Ma K, et al. Integrated demand-side management for multi-energy system based on non-cooperative game and multi-energy pricing. Sustainable Energy Grids Networks 2023;34:101047.
[12] Li K, Ye N, Li S, et al. Distributed collaborative operation strategies in multi-agent integrated energy system considering integrated demand response based on game theory. Energy 2023;273:127137.
[13] Yin X, Ye C, Ding Y, et al. Exploiting internet data centers as energy prosumers in integrated electricity-heat system. IEEE Trans Smart Grid 2022;14(1):167–82.
[14] Han O, Ding T, Mu C, et al. Coordinative optimization between multiple data center operators and a system operator based on two-level distributed scheduling algorithm. IEEE Internet Things J 2022;10(9):7517–27.
[15] Lv S, Chen S, Wei Z, et al. Incentivizing the Collaboration Between Travelers and Power-Traffic Network Operators: An Asymmetric Nash Bargaining Approach. IEEE Trans Sustainable Energy 2023;15(2):1328–40.
[16] Long X, Li Y, Li Y, et al. Collaborative response of data center coupled with hydrogen storage system for renewable energy absorption. IEEE Trans Sustainable Energy 2023;15(2):986–1000.
[17] Bian Y, Xie L, Ye J, et al. A new shared energy storage business model for data center clusters considering energy storage degradation. Renew Energy 2024;225:120283.
[18] Bian Y, Xie L, Ma L, et al. A novel two-stage energy sharing model for data center cluster considering integrated demand response of multiple loads. Appl Energy 2025;384:125454.
[19] Tian Q, Guo Q, Nojavan S, et al. Robust optimal energy management of data center equipped with multi-energy conversion technologies. J Clean Prod 2021;329:129616.
[20] Zárate-Miñano R, Cañas-Carretón M, Carrión M. Hybridization of existing solar PV farms considering wind power, batteries and electrolyzers. Int J Electr Power Energy Syst 2024;155:109557.
[21] Xia T, Hou Q, Zhang N, et al. Database Generation for Data-Driven Power System Security Assessment under Uncertainty. IEEE Trans Power Syst 2024;39(5):6168–82.
[22] Lian Y, Li Y, Zhao Y, et al. Robust multi-objective optimization for islanded data center microgrid operations. Appl Energy 2023;330:120344.
[23] Li X, Liu Y, Guo L, et al. Data-driven based uncertainty set modeling method for microgrid robust optimization with correlated wind power. CSEE J Power Energy Syst 2022;9(2):420–32.
[24] Lin J, Lin W, Lin W, et al. Thermal prediction for air-cooled data center using data driven-based model. Appl Therm Eng 2022;217:119207.
[25] Guo X, Yan F, Xue X, et al. QoS-aware data center network reconfiguration method based on deep reinforcement learning. Journal of Optical Communications and Networking 2021;13(5):94–107.
[26] Chung WK, Li Y, Ke CH, et al. Dynamic parallel flow algorithms with centralized scheduling for load balancing in cloud data center networks. IEEE Trans Cloud Comput 2021;11(1):1050–64.
[27] Chatterjee S, Misra S, Khan SU. Optimal data center scheduling for quality of service management in sensor-cloud. IEEE Trans Cloud Comput 2015;7(1):89–101.
[28] He Q, Lin Z, Chen H, et al. Bi-level optimization based two-stage market clearing model considering guaranteed accommodation of renewable energy generation. Prot Control Mod Power Syst 2022;7(3):1–13.
[29] Bian Y, Xie L, Ma L, et al. A novel two-stage energy sharing method for data center cluster considering 'Carbon-Green Certificate'coupling mechanism. Energy 2024;313:133991.
[30] Wang H, Li K, Zhang C, et al. Capacity and operation joint optimization for integrated energy system based on Nash bargaining game. Energy 2024;305:132032.
[31] Bertsimas D, Sim M. Robust discrete optimization and network flows. Math Program 2003;98(1):49–71.
[32] Velloso A, Street A, Pozo D, et al. Two-stage robust unit commitment for co-optimized electricity markets: An adaptive data-driven approach for scenario-based uncertainty sets. IEEE Trans Sustainable Energy 2019;11(2):958–69.
[33] Chen C, Li Y, Qiu W, et al. Cooperative-game-based day-ahead scheduling of local integrated energy systems with shared energy storage. IEEE Trans Sustainable Energy 2022;13(4):1994–2011.